Reduced Optimal Control with PDE Constraint
Model, Discretization, and First-Order Methods

Fachseminar (Mathematics Seminar)

March 15, 2026

Project in one slide

- **Goal:** find a control u (volumetric source) that drives a Poisson state y close to a desired profile y_d.
- **Constraint:** $-\Delta y = \chi_\omega\, u$ in Ω, $y = 0$ on $\partial\Omega$.
- **Cost (Tikhonov):**

$$J(y,u) = \tfrac{1}{2}\|y - y_d\|^2_{L^2(\Omega)} + \tfrac{\beta}{2}\|u\|^2_{L^2(\Omega)}, \qquad \beta > 0.$$

- **Approach:** P1 finite elements on a mesh family, state elimination $\Rightarrow$ reduced strictly convex quadratic $F(u)$.
- **Solvers:** CG on the stationarity system and first-order methods (GD $1/L$, BB, Nesterov with (m, L)).

Model problem

Domain. Polygonal $\Omega \subset \mathbb{R}^2$ (here $\Omega = [0,1]^2$). Control acts only on a tagged subdomain $\omega \subset \Omega$.

PDE constraint.

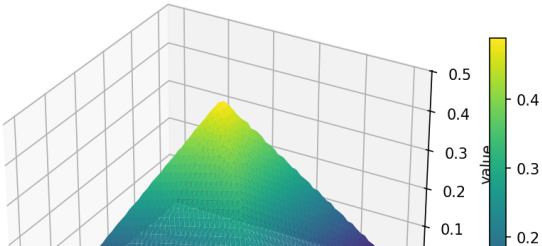
$$-\Delta y = \chi_\omega u \quad \text{in } \Omega, \qquad y = 0 \quad \text{on } \partial\Omega.$$

Spaces. $V = H_0^1(\Omega)$, $U = L^2(\Omega)$.

Target.

$$y_d(x) = 0.5 - \max\left(|x_1 - 0.5|, |x_2 - 0.5|\right) \in L^2(\Omega).$$

Target y_d (3D, h=0.02)



Weak formulation

Define

$$a(y, v) = \int_\Omega \nabla y \cdot \nabla v \, dx, \qquad b(u, v) = \int_\omega u \, v \, dx.$$

State: given $u \in L^2(\Omega)$, find $y \in V$ such that

$$a(y, v) = b(u, v) \quad \forall v \in V.$$

Cost:

$$J(y, u) = \tfrac{1}{2} \int_\Omega (y - y_d)^2 \, dx + \tfrac{\beta}{2} \int_\Omega u^2 \, dx.$$

Well-posedness: coercivity of $a(\cdot, \cdot)$ on $V \Rightarrow$ unique $y(u)$ (Lax–Milgram).

Optimality conditions (continuous)

Lagrangian

$$\mathcal{L}(y,u,p) = \tfrac{1}{2}\|y - y_d\|_{L^2}^2 + \tfrac{\beta}{2}\|u\|_{L^2}^2 + a(y,p) - b(u,p).$$

Stationarity yields:

$$\begin{aligned}
&\textbf{State:} && a(y,v) = b(u,v) && \forall v \in V,\\
&\textbf{Adjoint:} && a(v,p) = (y - y_d, v)_{L^2(\Omega)} && \forall v \in V,\\
&\textbf{Gradient:} && (\beta u + \chi_\omega p,\ w)_{L^2(\Omega)} = 0 && \forall w \in U.
\end{aligned}$$

Formally: $\beta u + \chi_\omega p = 0$ in Ω.

Finite element discretization (P1)

Let $V_h \subset H_0^1(\Omega)$ be the conforming P1 space with basis $(\varphi_i)_{i=1}^n$.

Assemble matrices

$$A_{ij} = \int_\Omega \nabla\varphi_j \cdot \nabla\varphi_i \, dx, \qquad M_{ij} = \int_\Omega \varphi_j \varphi_i \, dx,$$

$$B_{ij} = \int_\omega \varphi_j \varphi_i \, dx, \qquad (M_U)_{ij} = \int_\Omega \varphi_j \varphi_i \, dx.$$

Discrete state: $A\boldsymbol{y} = B\boldsymbol{u}$ (Dirichlet BCs imposed in the solve).
Discrete objective:

$$J(\boldsymbol{y}, \boldsymbol{u}) = \tfrac{1}{2}(\boldsymbol{y} - \boldsymbol{y}_d)^\top M(\boldsymbol{y} - \boldsymbol{y}_d) + \tfrac{\beta}{2}\boldsymbol{u}^\top M_U \boldsymbol{u}.$$

Reduced formulation

Eliminate state: $\boldsymbol{y}(\boldsymbol{u}) = A^{-1}B\boldsymbol{u}$. Define reduced functional

$$F(\boldsymbol{u}) = \tfrac{1}{2}(A^{-1}B\boldsymbol{u} - \boldsymbol{y}_d)^\top M(A^{-1}B\boldsymbol{u} - \boldsymbol{y}_d) + \tfrac{\beta}{2}\boldsymbol{u}^\top M_U \boldsymbol{u}.$$

Adjoint:

$$A\boldsymbol{p} = M\left(\boldsymbol{y}(\boldsymbol{u}) - \boldsymbol{y}_d\right).$$

U**-gradient (Riesz w.r.t.** $\langle a, b\rangle_U = a^\top M_U b$**):**

$$\nabla F(\boldsymbol{u}) = M_U^{-1}B^\top \boldsymbol{p} + \beta\boldsymbol{u}.$$

Quadratic structure and stationarity system

Expand F as an unconstrained QP:

$$F(\boldsymbol{u}) = \tfrac{1}{2}\boldsymbol{u}^\top Q\boldsymbol{u} + \boldsymbol{c}^\top \boldsymbol{u} + \text{const},$$

with

$$Q = B^\top A^{-1} M A^{-1} B + \beta M_U \quad \text{(SPD)}, \qquad \boldsymbol{c} = -B^\top A^{-1} M \boldsymbol{y}_d.$$

Optimality:

$$Q\boldsymbol{u}_* = B^\top A^{-1} M \boldsymbol{y}_d.$$

Interpretation: strictly convex $\Rightarrow$ unique minimizer; can solve by Krylov (CG) matrix-free via Hessian action.

Spectral viewpoint: L and m in the U-metric

In the U-inner product, the reduced Hessian is self-adjoint and SPD:

$$H\boldsymbol{v} = M_U^{-1} B^\top A^{-1} M A^{-1} B \boldsymbol{v} + \beta \boldsymbol{v} \quad \Leftrightarrow \quad H = M_U^{-1} Q.$$

Extreme eigenvalues are generalized eigenvalues of (Q, M_U):

$$Q\boldsymbol{v} = \lambda\, M_U \boldsymbol{v}.$$

Hence

$$L = \lambda_{\max}, \qquad m = \lambda_{\min}, \qquad \kappa = L/m.$$

Used for GD step $\alpha = 1/L$ and constant-parameter Nesterov.

Algorithms used

CG (linear solve). Solve SPD system $Q\boldsymbol{u} = \boldsymbol{b}$ with $\boldsymbol{b} = B^\top A^{-1}(M\boldsymbol{y}_d)$.

First-order methods on F (using U-gradient).

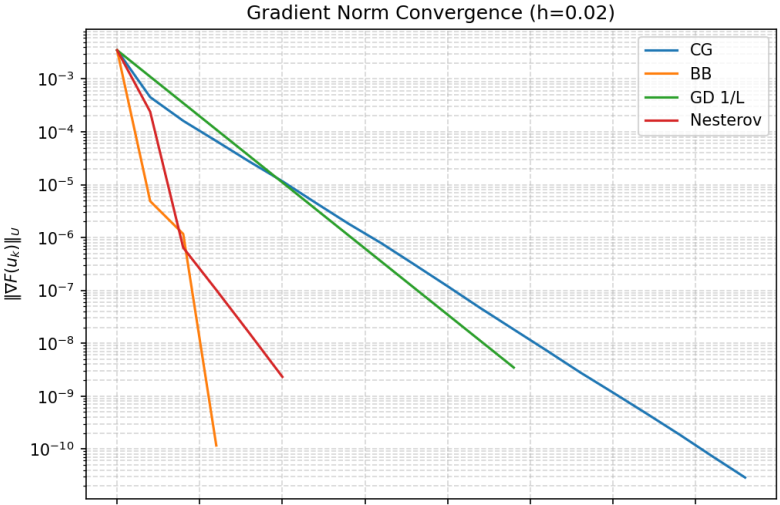
- GD with fixed step $\alpha = 1/L$.
- Barzilai–Borwein (alternating BB1/BB2) using secant information in $\langle \cdot, \cdot \rangle_U$.
- Nesterov acceleration for strongly convex case with parameters from (m, L).

Stopping criterion. $\|\nabla F(\boldsymbol{u}_k)\|_U \leq \texttt{tol}$ (and max iterations safeguard).

Representative numerical results ($h = 0.02$)

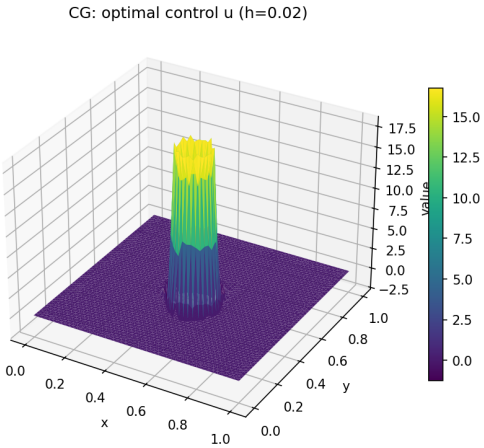
Parameters in experiments:

$$\beta = 10^{-3}, \qquad \omega \text{ tagged as physical region 2.}$$

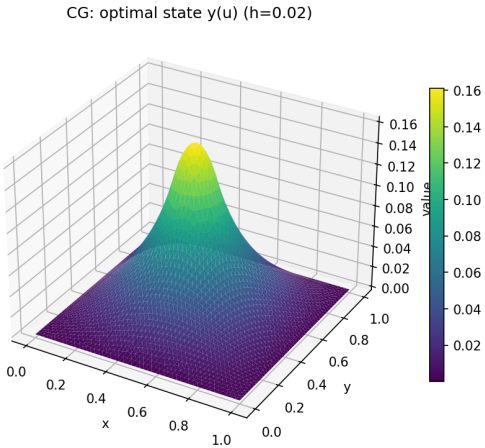


Optimized fields on representative mesh ($h = 0.02$)

Control u_* (CG)

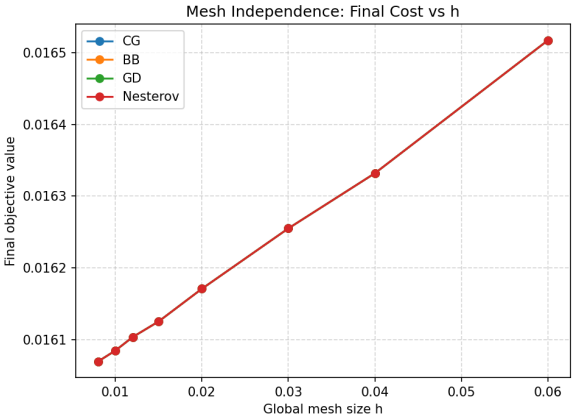


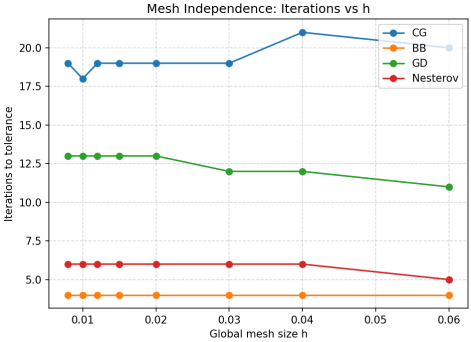
State $y(u_*)$ (CG)



Mesh-independence study

Meshes: $h \in \{0.06, 0.04, 0.03, 0.02, 0.015, 0.012, 0.01, 0.008\}$, pre-generated and loaded from disk.





- Objective and norms stabilize as $h \to 0$.
- Iteration counts stay largely insensitive to DOFs in this experiment.

Mesh-independence tables (from computed results)

Objective and norms (per mesh).

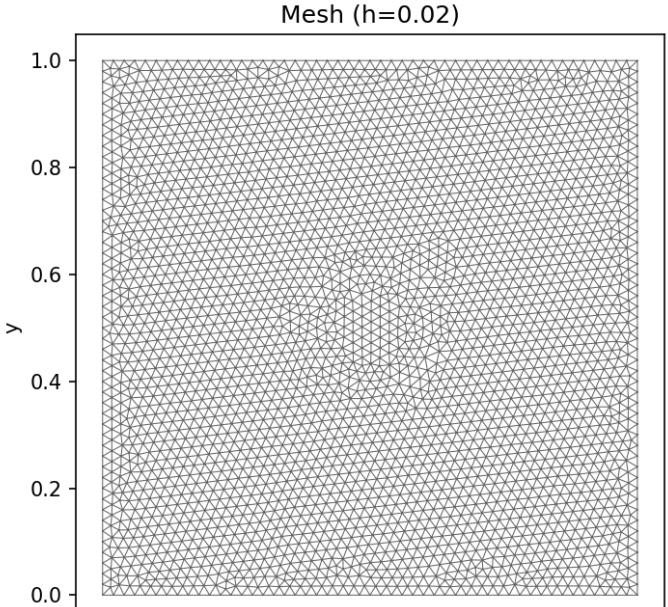
h	DOFs	$F(u_*^{(h)})$	$\|u_*^{(h)}\|_U$	$\|y(u_*^{(h)})\|_{L^2}$
0.06	402	0.0165171	2.549352	0.042344
0.04	820	0.0163319	2.607213	0.044949
0.03	1434	0.0162550	2.632511	0.046133
0.02	3093	0.0161708	2.654891	0.047219
0.015	5392	0.0161253	2.665356	0.047746
0.012	8438	0.0161035	2.671027	0.048028
0.01	11899	0.0160847	2.675266	0.048244
0.008	18497	0.0160695	2.678687	0.048419

Iterations (per mesh).

h	CG	BB	GD($1/L$)	Nesterov
0.06	20	4	11	5
0.04	21	4	12	6
0.03	19	4	12	6
0.02	19	4	13	6
0.015	19	4	13	6

Takeaways

- Reduced formulation turns PDE-constrained OC into an SPD quadratic problem in the control.
- The U-metric (mass matrix M_U) gives the natural gradient and convergence test.
- CG is highly efficient for the stationarity system; BB and Nesterov provide strong first-order baselines.
- Mesh-independence: optimal values stabilize; iteration counts are largely DOF-insensitive in these runs.

Implementation (high level): matrix-free Hessian action via state/adjoint solves; mesh family and plots generated by project scripts.

Backup: example mesh (optional)